7.2.1 From containers to Pods

As we discussed in the previous section, Pods are the smallest deployable units in Kubernetes. When moving from Docker to Kubernetes, we switch from managing containers to managing Pods.

A *Pod* is the smallest Kubernetes object, and it "represents a set of running containers" in a cluster. It's usually set up to run a single primary container (your application), but it can also run optional helper containers with additional features like logging, monitoring, or security (https://kubernetes.io/docs/reference/glossary).

A Pod is usually comprised of one container: the application instance. When that happens, it's not much different from working with containers directly. However, there are some scenarios where your application container needs to be deployed together with some *helper* containers that perhaps perform initialization tasks required by the application or add extra functionality such as logging. For example, Linkerd (a *service mesh*) adds its own container (a *sidecar*) to Pods to perform operations such as intercepting HTTP traffic and encrypting it to guarantee secure communication between all Pods via mTLS (mutual Transport Layer Security). Figure 7.3 illustrates single-container and multi-container Pods.